

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

MID SEMESTER EXAMINATION — EXAMINATION QUESTION PAPER

Subject: Operating Systems

Time: 1.5 Hours

Semester / Programme: 4th Semester

Full Marks: 20

Instructions: Answer Question No. 1 (Compulsory) and any THREE from Question No. 2 to 5. Figures in brackets indicate marks.

SECTION - A (Compulsory)

- 1(a)** Is context switching between threads in the same process less expensive than switching between threads in different processes? Justify your answer in one sentence. [1]
- 1(b)** A counting semaphore S is initialized to 12. Then 8 $wait()$ operations and 5 $signal()$ operations were completed on this semaphore. What would be the resulting value of S ? [1]
 $S_{final} = S_{initial} - (\text{number of } wait) + (\text{number of } signal)$
- 1(c)** Name any two CPU scheduling policies where preemption due to time quantum expiration does not occur. [1]
- 1(d)** Which of the following is a valid process state transition? (a) Ready $\rightarrow$ Ready (b) Ready $\rightarrow$ Waiting (c) Waiting $\rightarrow$ Ready (d) Waiting $\rightarrow$ Running [1]
- 1(e)** Describe the primary role of the medium-term scheduler in relation to system performance. [1]

SECTION - B (Answer any THREE)

Answer any THREE questions from Q.2 to Q.5. Each question carries 5 marks.

- Q.2.** (a) What is a system call? Explain the dual-mode operation of a computer system and how system calls facilitate the transition from user mode to kernel mode. [2.5]
- (b) Differentiate between user-level threads (ULT) and kernel-level threads (KLT) with respect to scheduling and kernel awareness. [2.5]
- Q.3.** (a) Consider the following set of processes, with the CPU burst time given in milliseconds: | Process | Burst Time | Arrival Time | | :--- | :--- | :--- | | P_1 | 6 | 0 | | P_2 | 8 | 1 | | P_3 | 7 | 2 | | P_4 | 3 | 3 | Calculate the average waiting time using the Preemptive Shortest Job First (SRTF) scheduling algorithm. [3]
 $Waiting\ Time = Completion\ Time - Arrival\ Time - Burst\ Time$
- (b) State the necessary conditions for a deadlock to occur according to Coffman. [2]
- Q.4.** (a) Explain Peterson's solution for the critical section problem of two processes. Prove that it satisfies the progress and bounded-waiting requirements. [3]
- (b) What is a binary semaphore? How does it differ from a counting semaphore? [2]

Q.5. (a) Explain contiguous memory allocation. Discuss the primary issues of internal and external fragmentation. [2.5]

(b) Given a paging system with a logical address space of 32 pages of 4KB each, mapped onto a physical memory of 1MB. Calculate the total number of bits in the logical address and the size of the page table entry assuming frame number field requires appropriate bits. [2.5]

Logical Address Bits = $\log_2$ (Total Pages) + $\log_2$ (Page Size)

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